

A sharp Pauli ℓ_1 bound for degree-one unitaries and a collision-free higher-degree criterion

Candidate partial result for arXiv:2405.10933

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Status. Candidate partial result, likely valid, pending human review. The full question remains open for fixed degrees $d \geq 2$.

1 Source question

Arunachalam, Dutt, Escudero Gutiérrez, and Palazuelos ask the following in Question 4 on page 6 of [1].

Is there a constant $C(d)$ such that

$$\sum_{x \in \{0,1,2,3\}^n} |\hat{U}(x)| \leq C(d)$$

for every n -qubit degree- d unitary U ?

Here $U = \sum_x \hat{U}(x) \sigma_x$ is the Pauli expansion and the degree is the largest number of nonidentity tensor factors in a Pauli string having a nonzero coefficient. We write

$$\mathcal{A}(U) := \sum_x |\hat{U}(x)|.$$

2 Proof intuition

At degree one, separate the identity coefficient from the traceless local piece on each qubit. Unitarity has a separate equation on every two-qubit support. That equation is a sum of two simple tensors, so it can vanish only when each local piece is a phase times a Hermitian matrix. Two active qubits must carry phases differing by $\pi/2$; three active qubits would impose three mutually incompatible phase differences. Thus a degree-one unitary is automatically a two-qubit junta. The remaining estimate is the sharp ℓ_1 – ℓ_2 inequality for two three-dimensional Pauli vectors.

This proves only the first nontrivial degree. Tensoring its extremizer gives an exponential lower bound on any possible constants in the full problem. For higher degree, a Hermitian dilation reduces arbitrary unitaries to Hermitian involutions at the cost of one degree. We also isolate a concrete mechanism that every difficult example must use: if the products of distinct commuting Pauli strings in the Fourier support do not collide, then the support is pairwise anticommuting and a sunflower argument gives an explicit dimension-free bound.

Consider the Pauli decomposition of an n -qubit unitary as follows:

$$U = \sum_{x \in \{0,1,2,3\}^n} \widehat{U}(x) \sigma_x,$$

where $\widehat{U}(x)$ are its Pauli coefficients. The degree of U is the minimum integer d such that $|x| > d$ implies $\widehat{U}(x) = 0$. Our learning model is similar to the one by Chen et al. [CNY23]. Given an unitary U , a learning algorithm is allowed to make *queries to U* (and to control- U) as follows: it can choose a state ρ , apply U to the state to obtain $U\rho U^*$, and measure $U\rho U^*$ in a chosen basis. From the measurement outcomes, the learner should output a classical description of an operator $\widetilde{U}$ that is close to U in the ℓ_2 -distance determined by the usual inner product for operators defined as $\langle U, V \rangle = \text{Tr}[U^*V]/2^n$.

Theorem 3. *Let U be an n -qubit degree- d unitary. There is an algorithm that (ε, δ) -learns U (in ℓ_2 -distance) using the unitary $U \exp(\widetilde{O}(d^2 + d \log(1/\varepsilon))) \cdot \log(1/\delta)$ many times.*

The proof of Theorem 3 follows the same structure as the one of Theorem 1, but now we learn the Pauli coefficients via the algorithm of Montanaro and Osborne [MO08], and the correctness of the algorithm relies on the non-commutative BH inequality of Volberg and Zhang [VZ23].

The BH inequality of Volberg and Zhang [VZ23] works for matrices with bounded operator norm, of which unitaries are a very special case. As argued by Montanaro and Osborne [MO08], matrices bounded on the operator norm are the quantum analogue of bounded functions $f : \{-1, 1\}^n \rightarrow [-1, 1]$, while unitaries are the analogue of Boolean functions⁴ $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$. These two families of classical functions differ a lot with respect to the BH inequalities: the best upper bound [DMP19] for the BH constant for bounded functions is $\exp(d^{1/2})$, while for Boolean functions of degree d one can even prove that the *usually much bigger* quantity $\sum_s |\widehat{f}(s)|$ is at most 2^{d-1} [O'D14, Exercise 1.11]. An open question is if this claim can be generalized to the quantum setting.

Question 4. *Is there a constant $C(d)$ such that $\sum_{x \in \{0,1,2,3\}^n} |\widehat{U}(x)| \leq C(d)$ for every n -qubit degree- d unitary U ?*

Figure 1: The definition and Question 4 in the source paper [1, p. 6].

3 Sharp degree-one theorem

We use the normalized Hilbert–Schmidt inner product $\langle A, B \rangle = 2^{-n} \text{Tr}(A^*B)$, for which the Pauli strings are orthonormal. Hence every unitary satisfies

$$\sum_x |\widehat{U}(x)|^2 = 1. \tag{1}$$

Lemma 1 (two-tensor dependence). *Let x, x' be nonzero vectors in a complex vector space E and let y, y' be nonzero vectors in a complex vector space F . If*

$$x' \otimes y + x \otimes y' = 0,$$

then there is a nonzero scalar λ such that $x' = \lambda x$ and $y' = -\lambda y$.

Proof. If x' were not in the span of x , there would be a linear functional φ on E with $\varphi(x) = 0$ and $\varphi(x') = 1$. Applying $\varphi \otimes \text{id}_F$ would give $y = 0$, a contradiction. Thus $x' = \lambda x$, and substitution gives $y' = -\lambda y$. $\square$

Theorem 1. *Let U be an n -qubit unitary of Pauli degree at most one. Then:*

1. U acts nontrivially on at most two qubits;

2. $\mathcal{A}(U) \leq \sqrt{6}$;

3. the constant $\sqrt{6}$ is attained.

Consequently the optimal constant in Question 4 at degree one is $C(1) = \sqrt{6}$.

Proof. Write

$$U = aI + \sum_{j=1}^n B_j^{(j)}, \quad B_j = b_{j1}X + b_{j2}Y + b_{j3}Z, \quad (2)$$

where the superscript means that B_j acts on the j th qubit and as the identity elsewhere. Call j active when $B_j \neq 0$.

Fix two distinct active qubits j, k . The component of $U^*U = I$ supported exactly on $\{j, k\}$ is

$$B_j^* \otimes B_k + B_j \otimes B_k^* = 0. \quad (3)$$

Lemma 1 gives a scalar λ_{jk} such that

$$B_j^* = \lambda_{jk} B_j, \quad B_k^* = -\lambda_{jk} B_k.$$

Taking adjoints twice shows $|\lambda_{jk}| = 1$. Thus every local piece having an active partner can be written

$$B_j = e^{i\theta_j} H_j$$

with H_j nonzero, Hermitian, and traceless. Equation (3) is then equivalent to

$$\theta_j - \theta_k \equiv \frac{\pi}{2} \pmod{\pi}. \quad (4)$$

There cannot be three active qubits: if j, k, ℓ were active, the equations for (j, k) and (j, ℓ) would give $\theta_k \equiv \theta_\ell \pmod{\pi}$, contradicting the equation for (k, ℓ) . This proves the junta assertion.

If no qubit is active, then $\mathcal{A}(U) = 1$. If exactly one is active, U is an embedded one-qubit unitary. It has at most four Pauli coefficients, so (1) and Cauchy–Schwarz give $\mathcal{A}(U) \leq 2$.

It remains to consider exactly two active qubits. Relabel them 1, 2. Write $B_j = e^{i\theta_j} H_j$ as above. Since a traceless Hermitian 2×2 matrix is $H_j = h_j \cdot (X, Y, Z)$ with $h_j \in \mathbb{R}^3$, we have

$$H_j^2 = r_j^2 I, \quad r_j := \|h_j\|_2.$$

The one-qubit components of $U^*U = I$ say

$$\Re(\bar{a}e^{i\theta_1}) = 0, \quad \Re(\bar{a}e^{i\theta_2}) = 0.$$

Because the two phases differ by $\pi/2$ modulo π , these two real equations force $a = 0$. The scalar component now gives

$$r_1^2 + r_2^2 = 1. \quad (5)$$

After a global phase and absorbing a sign into H_2 , the unitary therefore has the normal form

$$U = H_1 + iH_2, \quad H_j = h_j \cdot (X, Y, Z), \quad \|h_1\|_2^2 + \|h_2\|_2^2 = 1, \quad (6)$$

where the two terms act on distinct qubits. Hence

$$\mathcal{A}(U) = \|h_1\|_1 + \|h_2\|_1 \leq \sqrt{3}(r_1 + r_2) \leq \sqrt{6},$$

where both inequalities are Cauchy–Schwarz and the last uses (5).

For sharpness, put $Q = (X + Y + Z)/\sqrt{3}$. The Pauli anticommutation relations give $Q^2 = I$. Since $Q \otimes I$ and $I \otimes Q$ commute,

$$U_\star = \frac{Q \otimes I + iI \otimes Q}{\sqrt{2}} \quad (7)$$

is unitary. It has six nonzero degree-one Pauli coefficients, each of modulus $1/\sqrt{6}$, so $\mathcal{A}(U_\star) = \sqrt{6}$. $\square$

Corollary 1 (Hermitian subcase). *Every Hermitian degree-at-most-one unitary is a one-qubit junta and satisfies $\mathcal{A}(U) \leq \sqrt{3}$. The constant $\sqrt{3}$ is sharp.*

Proof. In the Hermitian case all B_j in (2) are Hermitian, so (3) becomes $2B_j \otimes B_k = 0$. Thus at most one qubit is active. A non-scalar Hermitian one-qubit unitary is traceless and has the form $h \cdot (X, Y, Z)$ for a real unit vector h . Its coefficient ℓ_1 norm is at most $\sqrt{3}$, with equality at Q above. $\square$

4 Necessary growth in higher degree

Corollary 2. *If constants $C(d)$ as in Question 4 exist, then for every positive integer d ,*

$$C(d) \geq 6^{d/2}.$$

Proof. Take $U_\star^{\otimes d}$ on $2d$ qubits. Pauli degree is additive under tensor products, while Pauli coefficient ℓ_1 norm is multiplicative. Thus this unitary has degree exactly d and

$$\mathcal{A}(U_\star^{\otimes d}) = \mathcal{A}(U_\star)^d = (\sqrt{6})^d.$$

$\square$

5 Hermitian dilation and a collision-free criterion

The next two statements do not settle Question 4 for $d \geq 2$, but they reduce the arbitrary-unitary problem to the Hermitian one and identify a necessary additive-combinatorial feature of any difficult support.

For a canonical Pauli string $P \in \{I, X, Y, Z\}^{\otimes n}$, write $[P]$ for its label modulo scalar phase. Thus, if P and Q commute, then $PQ = \pm R$ for the unique canonical Pauli string R with $[R] = [P][Q]$.

Lemma 2 (Hermitian dilation). *Suppose that every Hermitian unitary of Pauli degree at most $d+1$ has Pauli coefficient ℓ_1 norm at most $H(d+1)$. Then every unitary U of Pauli degree at most d satisfies*

$$\mathcal{A}(U) \leq H(d+1).$$

Proof. Write $U = A + iB$, where

$$A = \frac{U + U^*}{2}, \quad B = \frac{U - U^*}{2i}$$

are Hermitian, and introduce one ancilla qubit. The block matrix

$$F = \begin{pmatrix} 0 & U \\ U^* & 0 \end{pmatrix} = X \otimes A - Y \otimes B$$

is a Hermitian unitary of Pauli degree at most $d + 1$. If $\widehat{U}(P) = \alpha_P + i\beta_P$, then $\widehat{A}(P) = \alpha_P$ and $\widehat{B}(P) = \beta_P$. The X and Y parts of the dilation have disjoint Pauli supports, so

$$\mathcal{A}(F) = \sum_P (|\alpha_P| + |\beta_P|) \geq \sum_P |\alpha_P + i\beta_P| = \mathcal{A}(U).$$

The asserted bound follows. $\square$

Theorem 2 (collision-free support). *Let*

$$F = \sum_{P \in \mathcal{S}} a_P P, \quad a_P \in \mathbb{R} \setminus \{0\},$$

be a Hermitian unitary of Pauli degree at most d . Assume that the product labels of distinct commuting pairs in $\mathcal{S}$ are collision-free: if

$$P \neq Q, \quad R \neq S, \quad [P, Q] = [R, S] = 0, \quad [P][Q] = [R][S],$$

then $\{P, Q\} = \{R, S\}$. Then

$$\mathcal{A}(F) \leq \sqrt{M_d}, \quad M_d := 1 + \sum_{k=1}^d k! 3^{k^2+k}. \quad (8)$$

Consequently, if the support of the dilation in Lemma 2 is collision-free, then

$$\mathcal{A}(U) \leq \sqrt{M_{d+1}}.$$

Proof. Expanding $F^2 = I$ gives

$$F^2 = \sum_{P \in \mathcal{S}} a_P^2 I + \sum_{\{P, Q\} \subset \mathcal{S}} a_P a_Q \{P, Q\}.$$

An anticommuting pair contributes zero. A commuting pair contributes $\pm 2a_P a_Q R$ for the canonical Pauli string with label $[P][Q]$. Under the collision-free hypothesis, this nonzero coefficient cannot be cancelled by another pair. Since every nonidentity coefficient of F^2 vanishes, no two distinct members of $\mathcal{S}$ commute. Thus $\mathcal{S}$ is pairwise anticommuting.

It remains to bound the size of a pairwise anticommuting family of Pauli strings of weight at most d . Fix $k \geq 1$ and consider the members with support size exactly k . There are at most 3^k strings on any fixed support. The Erdős–Rado sunflower lemma [3] says that a family of more than $k!(r-1)^k$ distinct k -sets contains an r -petal sunflower. Take $r = 3^k + 1$. From such a sunflower choose one Pauli string on each support. Two of the chosen strings have the same Pauli labels on the common core, by pigeonhole. Their petals are disjoint, so these two strings commute, a contradiction. Hence there are at most $k!3^{k^2}$ distinct supports of size k , and therefore at most $k!3^{k^2+k}$ strings at that level. Adding the possible identity string proves $|\mathcal{S}| \leq M_d$.

Finally Parseval gives $\sum_{P \in \mathcal{S}} a_P^2 = 1$, and Cauchy–Schwarz gives $\mathcal{A}(F) \leq \sqrt{|\mathcal{S}|} \leq \sqrt{M_d}$. The last assertion follows from Lemma 2. $\square$

Why the collision condition is substantive. It is automatic at degree one after separating supports, but it can fail at degree two. If H_{ij} denotes the exact two-qubit component on $\{i, j\}$, then the four-qubit component of $F^2 = I$ contains

$$\{H_{ij}, H_{kl}\} + \{H_{ik}, H_{jl}\} + \{H_{il}, H_{jk}\} = 0.$$

The three perfect matchings of four vertices may therefore cancel. For example, if every edge term uses a fixed Pauli label at each endpoint, all three products have the same four-qubit Pauli label. This is exactly the collision phenomenon excluded in Theorem 2; controlling such structured collisions is the unresolved part of the full problem.

6 Verification, novelty, and limitations

The script `code/verify_extremizer.py` checks unitarity, Pauli degree, and the coefficient norm of (7), as well as coefficient norms of its first few tensor powers. This is a finite consistency check and is not used in the proof.

For novelty, searches were run through July 2026 for the exact wording of Question 4 and close variants involving “degree- d unitary”, “Pauli coefficients”, “Fourier ℓ_1 norm”, and “quantum Boolean degree one”. They located the source paper and the older Hermitian quantum-Boolean framework of Montanaro and Osborne [2], but no later paper stating this sharp classification for arbitrary degree-one unitaries. The latter paper’s quantum FKN results cover a closely related Hermitian setting; the source question, however, ranges over arbitrary unitaries, and the sharp extremizer above is non-Hermitian. Novelty confidence is moderate rather than definitive.

The result does not provide an unconditional upper bound for any fixed $d \geq 2$; the main question remains open there. Corollary 2 only constrains the growth of a possible positive answer, while Theorem 2 handles supports without repeated commuting products. The small-coefficient example

$$\epsilon X \otimes X + \sqrt{1 - \epsilon^2} Y \otimes I$$

from [2] also shows why a proof cannot rely on a positive lower bound for nonzero coefficients or influences.

Human-review recommendation. Review as a likely-valid sharp partial answer. The highest-value checks are Lemma 1, extraction of the two-qubit component (3), the phase normalization leading to (6), and the collision-free hypothesis in Theorem 2.

References

- [1] S. Arunachalam, A. Dutt, F. Escudero Gutiérrez, and C. Palazuelos, *Learning low-degree quantum objects*, arXiv:2405.10933, 2024.
- [2] A. Montanaro and T. J. Osborne, *Quantum boolean functions*, Chicago Journal of Theoretical Computer Science 2010(1), 2010; arXiv:0810.2435.
- [3] P. Erdős and R. Rado, *Intersection theorems for systems of sets*, Journal of the London Mathematical Society 35 (1960), 85–90.